

Emily V. Kerns

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Ph.D.	University of Wisconsin-Madison, Madison, WI Department of Integrative Biology <i>Thesis:</i> Evaluating mechanisms of resilience: from method comparisons across taxa to lab & natural experiments in threespine stickleback	2021 – 2026
B.S.	University of North Florida, Jacksonville, FL Biology, Concentration in Coastal and Marine Biology, Minor in Chemistry <i>Capstone:</i> Investigation of how selection is acting on a hybrid zone of <i>Fundulus grandis</i> and <i>F. heteroclitus</i>	2015 – 2018

RESEARCH EXPERIENCE

Cornell University, Ithaca, NY Advisor: Mal A. Ballinger Postdoctoral Research Associate Department of Ecology & Evolutionary Biology	2026 – Present
University of Wisconsin-Madison, Madison, WI Advisor: Jesse N. Weber Doctoral Research Associate Department of Integrative Biology	2021 – 2026
United States Forest Service, St. Regis, MT Biological Science Technician (Aquatic Systems) PacFish/InFish Biological Opinion Monitoring Program	2020
US Forest Service, AmeriCorps, Student Conservation Association (SCA), Tongass National Forest, AK Crew Leader / Biological Science Technician (Forestry) Tongass-Wide Young-Growth Study	2019
National Park Service, AmeriCorps, SCA, Redwood National Park, CA Biological Science Technician (Vegetation)	2019

Advisor: Matthew R. Gilg

Undergraduate Research Assistant

Department of Biology

PUBLICATIONS

[†]Corresponding author, ⁺Co-first author, ^{*}Undergraduate mentee

4. Bolnick D.I., Barrett R.D.H., Choi E., Eckert L., Hendry A.P., **Kerns E.V.**, Lind Å. J., Milligan-McClellan K., Peichel C.L., Sasser K.T., Thornton A.R., Wolf C., Steinel N.C., Weber J.N. (2026, September). Eco-evolutionary dynamics of infection and immunity in experimentally founded lake populations of threespine stickleback. *The American Naturalist*, 208(3). <https://doi.org/10.1086/741682>
3. **Kerns E.V.**[†], Engel S.⁺⁺, Codner P.A., Weber J.N. (2026, February). No evidence for quorum sensing during egg hatching in the cestode *Schistocephalus solidus*. *PeerJ* 14, e20667. <https://doi.org/10.7717/peerj.20667>
2. Hardy A.L., Gaither M.R., Lotterhos K.E., Greaves S., Cipolla K.J., **Kerns E.V.**, Trujillo A.P., Gilg M.R. (2025, February). Asymmetrical hybridization and environmental factors influence the spatial genetic structure of a killifish hybrid zone. *Evolution* 79(2), 232-248. <https://doi.org/10.1093/evolut/qpae160>
1. Gilg M.R., **Kerns E.V.**, Gutierrez-Bayona N.E., Kooyomjian C., Hinojosa N.A. (2022, March). Dynamic cohort analysis reveals fluctuating patterns of selection within a hybrid zone between the killifish *Fundulus heteroclitus* and *F. grandis*. *Evolutionary Biology* 49, 1-14. <https://doi.org/10.1007/s11692-021-09553-x>

In Review

2. **Kerns E.V.**[†], Weber J.N. Variable performance of widely used bisulfite sequencing methods and read mapping software for DNA methylation. *BioRxiv*. <https://doi.org/10.1101/2025.03.14.643302>. In review at *Ecology & Evolution*.
1. **Kerns E.V.**^{†+}, Sasser K.T.⁺, Weber J.N. Metabolic stability underpins thermal tolerance in benthic ecotypes of threespine stickleback (*Gasterosteus aculeatus*). *BioRxiv*. <https://doi.org/10.1101/2025.09.15.676414>. In review at *Proceedings of the Royal Society: B*.

Popular Press

1. **Kerns E.V.** Federal funding and my journey as a scientist. *Tampa Bay Newspapers*. Tampa Bay, FL. June 20, 2025. *McClintock Letters Initiative*.

PRESENTATIONS

Invited Seminars

2. **Kerns E.V.** (2025, November). Effects of inbreeding and introgression on iconic traits and resilience of McCloud River Redband trout. *Department of Ecology & Evolutionary Biology Seminar Series*. University of California Los Angeles. Los Angeles, CA.
1. **Kerns E.V.** (2025, October). How to connect phenotypic plasticity, epigenetics, and natural selection? Insights from thermal tolerance and DNA methylation experiments in threespine stickleback. *Evolution Seminar Series*. University of Wisconsin-Madison. Madison, WI.

Conference Presentations

+Co-first author presentation

6. **Kerns E.V.**, Yu C., Weber J.N. (2026, June). Testing a novel framework for connecting phenotypic plasticity, epigenetics, and natural selection using experiments in threespine stickleback. *Evolution*. Cleveland, OH.
5. **Kerns E.V.**, Weber J.N. (2024, July). How do commonly used methods capture variation in DNA methylation of threespine stickleback (*Gasterosteus aculeatus*)? *Evolution*. Montreal, Canada.
4. **Kerns E.V.**, Weber J.N. (2024, July). How do commonly used methods capture variation in DNA methylation of threespine stickleback (*Gasterosteus aculeatus*)? *Alaska Stickleback Project Retreat*. Montreal, Canada.
3. **Kerns E.V.**, Weber J.N. (2023, April). Does DNA methylation mediate flexible responses in threespine stickleback? *Alaska Stickleback Project Retreat*. Montreal, Canada.
2. Sasser K.T.⁺, **Kerns E.V.**⁺, Weber J.N. (2022, February). Fish in hot water: Thermal tolerance and interacting stressors. *Alaska Stickleback Project Retreat*. Virtual.
1. Gilg M.R., **Kerns E.V.**, Gutierrez-Bayona N.E., Kooyomjian C., Hinojosa N.A. (2019, July). Dynamic cohort analysis reveals evidence of directional selection in a killifish hybrid zone. *Joint Meeting of Ichthyologists and Herpetologists*. Snowbird, UT.

Poster Presentations

6. **Kerns E.V.**, Yu C., Sasser K.T., Weber J.N. (2026, February). How to connect phenotypic plasticity, epigenetics, and natural selection? Insights from thermal tolerance and DNA methylation experiments in threespine stickleback. *UW-Madison Joint Biological Sciences Poster Session*. Madison, WI.
5. **Kerns E.V.**, Yu C., Sasser K.T., Weber J.N. (2025, October). How to connect phenotypic plasticity, epigenetics, and natural selection? Insights from thermal tolerance and DNA methylation experiments in threespine stickleback. *Wisconsin Epigenetics Symposium*. Madison, WI.

4. **Kerns E.V.**, Sasser T., Weber J.N. (2022, October). How do DNA methylation and gene expression mediate plasticity of threespine stickleback (*Gasterosteus aculeatus*)? *Wisconsin Epigenetics Symposium*. Madison, WI.
3. **Kerns E.V.**, Sasser T., Weber J.N. (2022, June). How do DNA methylation and gene expression mediate plasticity of threespine stickleback (*Gasterosteus aculeatus*)? *Evolution*. Cleveland, OH.
2. **Kerns E.V.**, Cipolla K., Gonzalez D., Gilg M.R. (2019, July). Landscape genetics of a *Fundulus spp.* hybrid zone in northeast Florida. *Joint Meeting of Ichthyologists and Herpetologists*. Snowbird, UT.
1. **Kerns E.V.**, Gilg M.R. (2018, February). Hybrid zone analysis of *Fundulus grandis* and *F. heteroclitus*. *Florida Undergraduate Research Conference*. Cocoa, FL.

Guest Lectures

2. Evolutionary Medicine. (2025, April). University of Wisconsin-Madison Evolutionary Biology 410, taught by Dr. Jesse N. Weber. Madison, WI.
1. Evolutionary Medicine. (2023, May). University of Wisconsin-Madison Evolutionary Biology 410, taught by Dr. Jesse N. Weber. Madison, WI.

HONORS AND AWARDS

Research Awards: \$14,520

UCLA La Kretz Center Postdoctoral Fellowship in California Conservation Science. <i>Finalist</i>	2026
American Society of Naturalists Graduate Student Travel Award. \$500	2026
UW-Madison Integrative Biology JDD Travel Award. \$800, \$900	2024, 2026
Society for the Study of Evolution, Graduate Research Excellence Grant, Rosemary Grant Advanced Award. \$3,500	2024
UW-Madison Integrative Biology Graduate Summer Research Award. \$7,520	2022
National Science Foundation Graduate Research Fellowship Program. <i>Honorable Mention</i>	2022
UNF Undergraduate Research Grant. \$1,300	2017

Teaching & Service Awards: \$4,200

William C. Burns & Lemuel F. Fraser Teaching Enrichment Scholarship. \$1,000	2023
AmeriCorps Education Award. \$1,600, \$1,600	2019, 2019

Unfunded Applications: \$382,330

NIH R01 Administrative Supplement under Climate Change and Health Notice of Special Interest. PI: Daniel Bolnick. Co-authors and Supporting Personnel: Benjamin Flanagan, Emily Kerns. \$382,330	2023
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TEACHING EXPERIENCE

University of Wisconsin-Madison

Madison, WI

Center for Integration of Research, Teaching, & Learning

2025 Delta Program Teaching & Learning Practitioner Badge: Does integrating periodic utility value interventions in an introductory biology course increase undergraduate students' intrinsic interest in evolution, ecology, and animal biology?

Expeditions in Evidence-Based Teaching and Learning 2024

Teaching in Science and Engineering: The College Classroom 2023

Research Mentor Training 2022

Teaching Assistant

Animal Biology Laboratory 2024, 2025, 2026

Evolutionary Biology Discussion 2023, 2025

Introductory Biology 2 Laboratory 2021

University of North Florida

Jacksonville, FL

Supplemental Instructor

General Biology 2 2018

SERVICE AND OUTREACH

University of Wisconsin-Madison

Madison, WI

Wisconsin Evolution

Evolution Coordinating Committee 2024 – 2026

University Community Engagement Committee 2025 – 2026

Evolution Seminar Series Committee 2024 – 2025

Department of Integrative Biology

Graduate Student Organization Outreach Coordinator 2024 – 2026

UW Science Expeditions: Exploration Station 2025, 2026

>2,000 community members reached

Wisconsin Science Festival: Science on the Square 2024, 2025

>1,000 community members reached

Wisconsin Science Festival: Community Expo 2024

>600 community members reached

Graduate Program Manager Hiring Committee 2025

Undergraduate Research Mentees:

Anna Chung January 2025 – May 2026

Claire An November 2024 – May 2026

Molly O'Neill May – September 2025

Daisy Hester
Sara Engel

June – September 2024
September 2022 – September 2023

Badger Buddies, Biological Interactions Summer Research Program (WISCIENCE)

Professional Development Mentor

2022

Co-Reviewer:

Journals

eLife, Nature Ecology & Evolution, Molecular Ecology Resources

Funding agencies

French National Research Agency (2025)

Alaska Stickleback Project

Virtual

Graduate Student and Postdoctoral Presentations Co-Organizer

2023 – 2024

PubNavigator: Science Communication Platform

Virtual

Associate Editor

2020 – 2022

PROFESSIONAL AFFILIATIONS

American Society of Naturalists

2024 – Present

Society for the Study of Evolution

2022 – Present

SKILLS, WORKSHOPS, AND CERTIFICATIONS

Proficient in R, Python, Linux, Bash, & HTCondor Software Suite for high throughput computing

Introduction to Equity-Minded Mentorship. *Equity in Graduate Education*

2024

Resource Center

A Practical Approach to Project Management. *UW-Madison Graduate*

2023

School Office of Professional Development

Implicit Bias. *Fair Play Graduate Student and Postdoc Workshop*

2022

Wilderness First Aid. *Wilderness Medical Associates International*

2019

Shoreline Survival Training. *US Forest Service*

2019

Boating Safety Course, Florida Lifetime Certification. *BoatU.S Foundation.*

2018